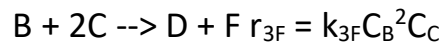
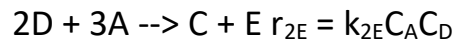


E09 Multiple Reactions in an Isothermal PFR

The following reactions are carried out in the gas phase in an isothermal PFR:



Where $k_{1D} = 0.25 \text{ L/mol} \cdot \text{min}$, $k_{2E} = 0.1 \text{ L/mol} \cdot \text{min}$, $k_{3F} = 5 \text{ L}^2/\text{mol}^2 \cdot \text{min}$, $C_{T0} = 0.4 \text{ mol/L}$, $V = 500 \text{ L}$, $y_{A0} = y_{B0}$, $u_0 = 100 \text{ L/min}$. Plot the species concentrations as functions of PFR volume.

Decision Guide: Multiple Reactions in an Isothermal PFR

Goal: Determine the concentration profiles of: $CA(V)$, $CB(V)$, $CC(V)$, $CD(V)$, $CE(V)$, $CF(V)$

throughout a plug-flow reactor carrying out multiple simultaneous reactions.

Step 1: Identify the Type of Problem

Ask yourself:

What reactor?

→ PFR

Multiple reactions?

→ Yes

Isothermal?

→ Yes

Need an energy balance?

→ No

Because the reactor temperature is constant, only species balances are required.

Step 2: List All Species

Identify every species appearing in the reaction network: A, B, C, D, E, F

These become the dependent variables of the model.

Step 3: Identify the Reactions

Reaction 1: $A + 2B \rightarrow C + D$

Reaction 2: $2D + 3A \rightarrow C + E$

Reaction 3: $B + 2C \rightarrow D + F$

Notice that several species participate in more than one reaction.

For example:

C is produced in Reactions 1 and 2 and consumed in Reaction 3

D is produced in Reactions 1 and 3 and consumed in Reaction 2

Step 4: Calculate the Reaction Rates

Use the rate laws provided in the problem statement.

Reaction 1: $r_1 = k_1 CA CB$

Reaction 2: $r_2 = k_2 CA CD$

Reaction 3: $r_3 = k_3 CB^2 CC$

where:

$k_1 = 0.25 \text{ L/mol}\cdot\text{min}$

$k_2 = 0.10 \text{ L/mol}\cdot\text{min}$

$k_3 = 5 \text{ L}^2/\text{mol}^2\cdot\text{min}$

Step 5: Construct a Species-Accounting Table

Determine how each reaction affects each species.

Species A:

Consumed in: Reaction 1, Reaction 2

Species B

Consumed in: Reaction 1, Reaction 3

Species C

Produced in: Reaction 1, Reaction 2

Consumed in: Reaction 3

Species D

Produced in: Reaction 1, Reaction 3

Consumed in: Reaction 2

Species E

Produced in: Reaction 2

Species F

Produced in: Reaction 3

Step 6: Write the Species Rates

The PFR balances require species rates, not reaction rates. For each species, combine the contributions from all reactions using stoichiometry. For example:

Species A

Rate of A = Consumption by Reaction 1 + Consumption by Reaction 2

Species B

Rate of B = Consumption by Reaction 1 + Consumption by Reaction 3

Species C

Rate of C = Production by Reaction 1 + Production by Reaction 2 – Consumption by Reaction 3

Species D

Rate of D = Production by Reaction 1 + Production by Reaction 3 – Consumption by Reaction 2

Species E

Rate of E = Production by Reaction 2

Species F

Rate of F = Production by Reaction 3

Use these relationships to write: r_A , r_B , r_C , r_D , r_E , r_F

in terms of the underlying reaction rates r_1 , r_2 , and r_3 .

Why Is This Step Necessary?

The kinetics provide:

Reaction Rates (r_1 , r_2 , r_3) but the PFR mole balances require species rates (r_A , r_B , r_C , r_D , r_E , r_F). Therefore, the reaction rates must be combined to determine the net rate of production or consumption of each species.

Workflow

Reaction Rates

↓

Species Rates

↓

Species Mole Balances

↓

Flowrate Profiles

↓

Concentration Profiles

Step 7: Write the PFR Mole Balances

For each species:

$$\frac{dF_i}{dV} = r_i$$

Use one balance for: A, B, C, D, E, F

This produces six coupled differential equations.

Step 8: Determine the Inlet Conditions

The problem provides:

$C_{T0} = 0.4 \text{ mol/L}$

$y_{A0} = y_{B0}$. Because only A and B are fed and their mole fractions are equal: $y_{A0} = 0.5$, $y_{B0} = 0.5$.

Therefore: $CA_0 = y_{A0} C_{T0} = 0.20 \text{ mol/L}$, $CB_0 = y_{B0} C_{T0} = 0.20 \text{ mol/L}$, and $CC_0 = 0 = CD_0 = CE_0 = CF_0$.

Step 9: Convert Concentrations and Flowrates

The rate laws require concentrations. At each point in the reactor:

$$C_i = F_i / v$$

where: C_i = concentration of species i , F_i = molar flowrate of species i , v = volumetric flowrate

Use the current species flowrates to calculate concentrations before evaluating the reaction rates.

Why Is This Step Necessary?

The PFR mole balances are usually written in terms of molar flowrates: F_A , F_B , F_C , F_D , F_E , F_F

However, the rate laws are written in terms of concentrations: C_A , C_B , C_C , C_D , C_E , C_F

Therefore, we must convert between flowrates and concentrations throughout the reactor.

What Happens During the Numerical Solution?

At each integration step:

Current Flowrates

↓

Calculate Concentrations

↓

Evaluate Reaction Rates

↓

Calculate Species Rates

↓

Update Flowrates

This process repeats until the reactor outlet is reached.

Step 10: Count Equations and Unknowns

Unknowns: F_A , F_B , F_C , F_D , F_E , F_F

Equations: Species A Balance, Species B Balance, Species C Balance, Species D Balance, Species E Balance, Species F Balance

The model is now complete.

Step 11: Solve Numerically

Integrate the six coupled PFR mole balances from $V = 0$ to $V = 500$ L using a numerical ODE solver such as Python `solve_ivp`.

Step 12: Plot the Concentration Profiles

Generate: C_A versus V , C_B versus V , C_C versus V , C_D versus V , C_E versus V , C_F versus V

to visualize how the reaction network evolves throughout the reactor.

Final Answer Checklist

- ✓ Identified all six species.
- ✓ Calculated all reaction rates.
- ✓ Constructed a species-accounting table.
- ✓ Converted reaction rates into species rates.
- ✓ Wrote six PFR mole balances.
- ✓ Applied inlet conditions correctly.
- ✓ Solved the coupled ODE system.
- ✓ Generated concentration profiles.

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (16) PFR Mole Balance

Eq. (47) Relative Rates

Key Insight

Reaction Rates



Species Rates



Species Balances



Concentration Profiles

In multiple-reaction systems, species are often produced in one reaction and consumed in another. As a result, concentration profiles frequently pass through maxima or minima rather than changing monotonically. The key step is converting the reaction rates into the net rate of production or consumption for each species.